AGRIPROVE
SOIL SAMPLING PROTOCOL

STANDARD OPERATING PROCEDURE

OPS006

Revision history

Revision	Author/Reviewer	Date	Remarks
0.1	Brendan Fallon	21/08/2020	Working Draft
0.2	Kieren Whittock	30/11/2020	Draft
0.3	Kate Carmichael	25/11/2021	Draft
1.0	Kieren Whittock	20/09/2022	Revision to Remove Quoting Process
1.1	Kieren Whittock	15/02/2023	Revision 1
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2.0	Kieren Whittock	21/03/2024	Publish
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2.2	Kieren Whittock	26/07/2024	Update for IS9001; Publish
2.3	Kieren Whittock	08/05/2025	QuickCapture; contractor onboarding

Contents

1 SCOPE

Physical soil sampling is a core component of the operation of AgriProve's soil carbon projects. It forms the basis of estimating Soil Organic Carbon stocks, and it is imperative that this procedure be followed to ensure project compliance.

This procedure has been developed to be compliant with the *Carbon Credits (Carbon Farming Initiative) Act 2011* (CFI Act), the *Carbon Credits (Carbon Farming Initiative) Regulations 2011*, the *Carbon Credits (Carbon Farming Initiative – Estimation of Soil Organic Carbon Sequestration using Measurement and Models) Methodology Determination 2021* (the Method), the *The Explanatory Statement - Authorised Version registered 02/12/2021 to F2021L01696* and *Supplement to the Carbon Credits Carbon Farming initiative – Estimation of Soil Organic Carbon Sequestration Using Measurement and Models Methodology Determination 2021* (the Supplement).

This procedure operationalises the sampling requirements as specified in the Supplement as listed above, however it is imperative that all personnel following this procedure have read an understood the requirements of the Supplement and how it interacts with this document.

2 RELATIONSHIPS

This Standard Operations Procedure is to be implemented in accordance with AgriProve's Quality Management System. AgriProve's Quality Policy will be provided during contractor onboarding and all third-party sampling contractors must agree to operate under the principles outlined in this policy.

3 REQUIREMENTS

It is a requirement that this procedure be read and understood prior to the collection of soil samples.

Access to the following systems is a minimum requirement for implementing this procedure. AgriProve's GIS administrator will issue login credentials prior to mobilisation:

- ArcGIS Field Maps
- ArcGIS Survey 123
- ArcGIS QuickCapture

4 SAFETY AND BIOSECURITY

Safety of all personnel is paramount. It is the responsibility of the employees or contractors undertaking the soil sampling to ensure that work is completed safely and in accordance with both AgriProve policies and procedures, and relevant state-based safe work legislation.

Under no circumstances should sample collection be attempted if it is deemed unsafe to do so.

The following processes are required to be completed prior to commencing work. The Contractor Onboarding Checklist provided as an Appendix to this document should be completed in full prior to commencing work.

1. All third part contractors must have a Work Health and Safety Policy and be able to provide this to AgriProve on request.

2. All third part contractors must hold appropriate workers compensation, professional indemnity, and public liability insurances and be able to provide Certificates of Currency prior to mobilisation.

3. For each project, the employees/contractors undertaking the work should complete a thorough review of the BYD information provided and be satisfied that they are aware of the location of utilities, noting that often the spatial accuracy of the assets is poorly constrained. AgriProve completes a BYD on all projects and integrates this data to sampling plans, however this data will be supplied to all contractors, and all contractors should satisfy themselves that AgriProve have adequately addressed any risks prior to sampling.

4. For each project a briefing with the landholder/farm manager should be completed to discuss access, and any site-specific risks or specific biosecurity requirements that may be present. AgriProve requires a record of this briefing. If an individual contractor does not have an existing template, AgriProve can supply a standard template as part of the contractor onboarding process.

5. Vehicle and equipment pre-start checks should be completed on a daily basis and retained. While AgriProve does not require these to be provided as standard, AgriProve reserves the right to request copies and proof of completion at any time.

6. On completion of each project, it is imperative that all equipment is thoroughly cleaned and washed down to prevent the spread of pests and disease between properties and that a record of this washdown is completed. AgriProve can provide a template if the contractor does not currently maintain records.

7. If there are any site-specific risks or tasks that are identified that are outside normal day to day operations, a risk assessment should be completed to determine whether the task can be safely completed. All contractors must have appropriate risk assessment procedures in place as part of their Work Health and Safety Policy.

8. It is imperative that all contractors have adequate communications available to them at all times, and also have measures in place to manage communications in areas of no mobile reception (satellite phones, Starlink etc).

9. A daily update should be supplied on progress to the Head of Operations (note this can be informal via phonecall or SMS).

All contractors should perform their work in a respectful manner whilst on-farm, and ensure all work is completed in a timely and professional manner, while minimising ground disturbance and disruption to farming operations. It is imperative that all sampling sites are left free of rubbish and in a safe condition.

5 ARCGIS FIELD MAPS AND SURVEY 123 CONFIGURATION

All sampling completed on AgriProve projects is to be completed using ArcGIS Field Maps and Survey 123 for in-field data acquisition. This ensures data quality and integrity and is critical for downstream data systems and applications to work efficiently.

Prior to mobilising for sampling, ensure that the ArcGIS and Survey 123 applications are downloaded to your device from the appropriate app store. You will be assigned a specific ArcGIS username and password to login to these applications by the GIS Administrator.

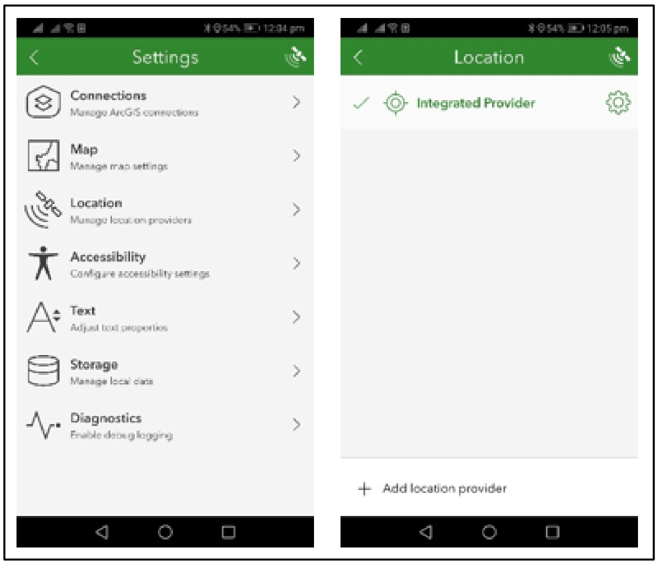
To complete set up:

1. Open both applications and login with the supplied credentials.

2. Within Survey 123, navigate to '*Download Surveys*' and download the *AgriProve_Soil_Sampling_Survey* form.

3. Establish a Bluetooth connection with the GPS system being used.

4. Open *Settings* and ensure that *Location* is set to the GPS system in use. **If the location is *Integrated Provider* this needs to be changed as this is the device GPS and will not provide location information to a high enough level of accuracy.**

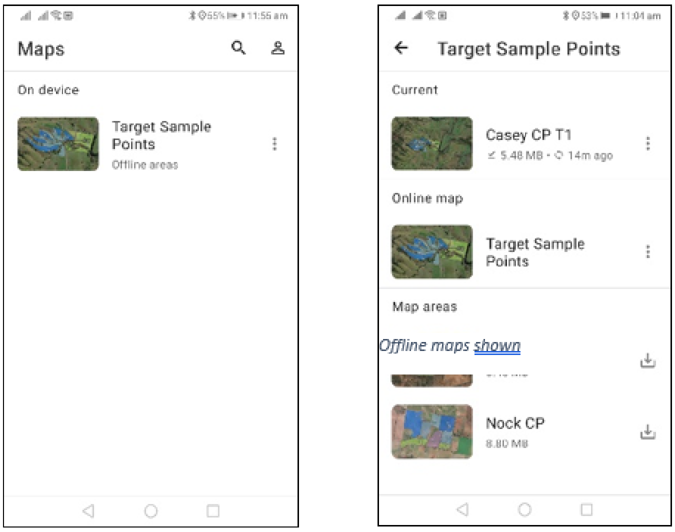


5. Leave all other settings as default unless instructed otherwise. Close Survey 123 and open Field Maps.

6. Click on the *Profile* button and in *Location* click on *Provider.* You will see the GPS device listed. Select your location profile using GDA2020 – 7844.

7. Under *General Settings* ensure that the units are set to metres. Exit *Profile.*

8. Download the relevant soil sampling map (there should only be one map available). In the *Offline Map* list, download the relevant project map.



These steps should all be completed and tested prior to mobilising to the field to ensure there are no unexpected issues that delay sampling.

6 SAMPLING PROTOCOL

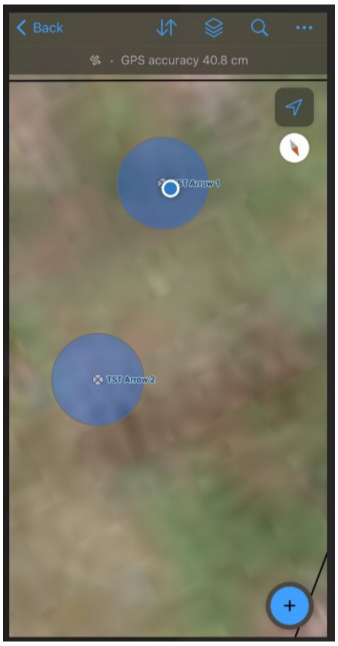
6.1 LOCATING SAMPLING LOCATIONS

Part C 1.0 (1) of the Supplement states that it is a requirement that a GNSS device with a minimum accuracy of +/- four metres is used to locate intended sampling locations. AgriProve field teams currently use an Arrow 100 Series GNSS receiver for spatial positioning, with a 60cm accuracy. Please check with the relevant AgriProve project manager if you are unsure as to whether the GPS device being used fits these criteria.

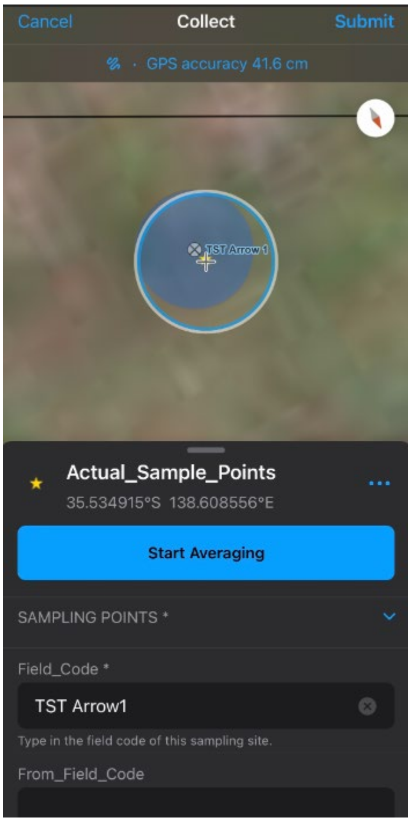
1. Ensure that your GPS system is connected to your device and configured with Field Maps and Survey 123 as detailed above.

2. Ensure Field Maps is open and the relevant map selected. Note that depending on availability of mobile data coverage, it may be preferable to work in an offline map.

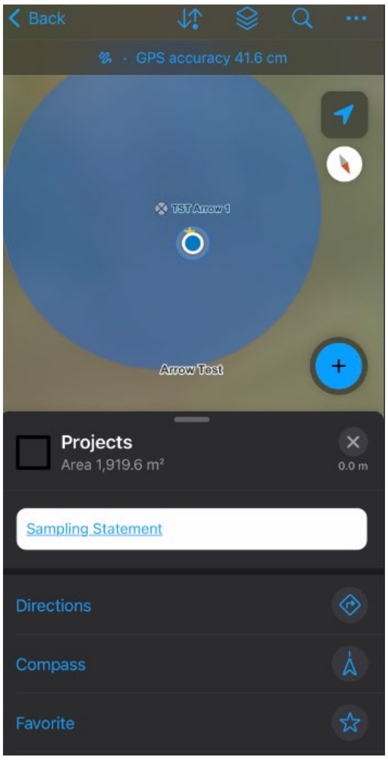
3. Your location will appear on the map as a small blue circle that will move as you change location. The target sampling points will appear on the map with a 4m wide blue buffer zone to assist in navigation. Care should be taken to navigate as close as possible to the target sampling location in the centre of the blue buffer zone. **Note that under no circumstances should a sample be taken outside of the 4m buffer zone.**



4. Prior to collecting the sample, ensure that the GPS accuracy, displayed at the top of the screen in Field Maps, is **less than 1m**. Once you are satisfied with your location, tap the sample point in Field Maps, and then tap the resulting *'Start Averaging'* button. Enter the Field Code as displayed on the map in the *Field_Code* dropdown.



5. Tap on the project name and select the relevant Field Code in the dropdown list. The Survey 123 form will open automatically. Begin completing the form. **Also ensure that the location is captured in Field Maps by tapping the blue circle in the bottom right of the screen. This ensures data redundancy and traceability in data collection.**



6.2 COMPLETING THE SURVEY 123 FORM

It is imperative that the Survey 123 form is filled out correctly, as this has been specifically designed to capture all the in-field information required for a compliant soil sampling round.

1. Answer the questions in the form, noting that all fields with a red asterisk are required fields. If the sample being collected is **not** a reserve point, select '*Yes*'.

2. Enter the project name and field code will automatically populate.

3. If your location is exactly where the sample is required to be extracted and the location is safely accessible, select '*Yes*' under '*Is this sample point location safe and accessible?'*

    a. **If the answer to this question is 'No' it is imperative that photographic evidence showing why the sample could not be extracted is acquired under the *Location of Record* section in Survey 123.**

    b. If the sample is able to be extracted but with an offset applied, follow the instructions for offsetting provided in the form, and ensure photographic evidence is taken of the obstruction, and the actual sampled location.

4. **For every sample it is imperative that the diameter of the cutting shoe is checked and confirmed. The diameter of the shoe may change as the cutting shoe wears.**

5. If there is no change to the cutting shoe diameter select '*No*' under '*Check cutting shoe and equipment*'. If there are any adjustment required to any equipment, select '*Yes*' and note the adjustments made.

6. Based on the checks made above, enter the **internal cutting shoe diameter** in centimetres under '*Extraction core diameter'*. This must not be less than 3.8cm.

7. Enter the '*Target soil push depth'* in centimetres. **This should always be 100cm unless otherwise specified in writing by AgriProve Operations**.

8. Extract the soil core, ensuring that no lubricants are used, and only water has been used to wash equipment between samples.

9. Enter the '*Actual push depth'* in centimetres. For the avoidance of doubt, this is the depth that the probe has been pushed into the ground, and not the actual sample length.

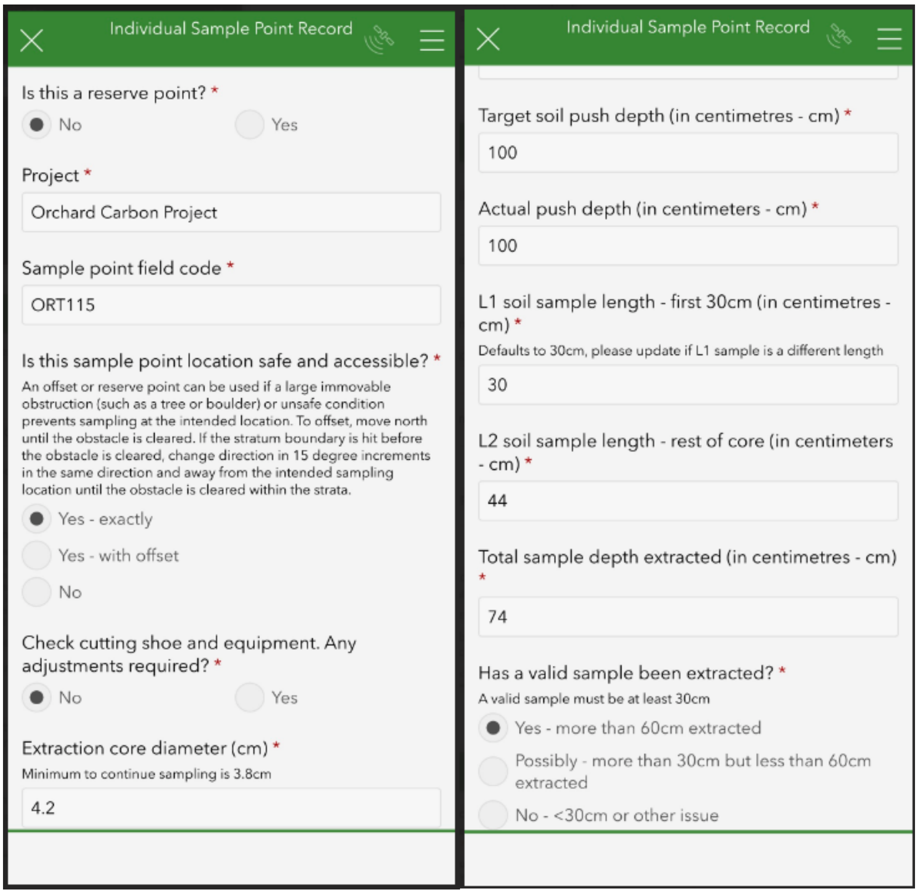
10. Extract the core from the tube and separate the L1 (0-30cm) from the L2 (30cm +). Ensure that the length of each layer is recorded correctly in the relevant fields in the Survey123 form. **Care must be taken to ensure these measurements are as accurate as possible as small errors can have significant impact on calculations for total carbon stocks.**

11. It is imperative the L1 layer is 30cm exactly and must not be less than 30cm. **No adjustments should be made to the sample lengths based on the compression of the core.**

12. Calculate the total length (L1 + L2) and add the total core length to the '*Total sample depth extracted'* field in centimetres.

13. If the total core length is > 60cm, proceed to complete the rest of the form, and collect the samples.

    a. If >30cm and <60cm total sample length has been obtained, consider a second attempt at extracting the core from the same location (move forward 20cm for example).
    b. If <30cm has been extracted, consider a second attempt from the same location. If >30cm of total core length is still not obtained, consider using the offsetting procedure to attempt to obtain a compliant sample (within the 4m spatial buffer). If this is still not successful, utilise a reserve point.


Individual Sample Point Record

Is this a reserve point? *
No Yes

Project *
Orchard Carbon Project

Sample point field code *
ORT115

Is this sample point location safe and accessible? *
An offset or reserve point can be used if a large immovable obstruction (such as a tree or boulder) or unsafe condition prevents sampling at the intended location. To offset, move north until the obstacle is cleared. If the stratum boundary is hit before the obstacle is cleared, change direction in 15 degree increments in the same direction and away from the intended sampling location until the obstacle is cleared within the strata.
Yes - exactly
Yes - with offset
No

Check cutting shoe and equipment. Any adjustments required? *
No Yes

Extraction core diameter (cm) *
Minimum to continue sampling is 3.8cm
4.2

Individual Sample Point Record

Target soil push depth (in centimetres - cm) *
100

Actual push depth (in centimeters - cm) *
100

L1 soil sample length - first 30cm (in centimetres - cm) *
Defaults to 30cm, please update if L1 sample is a different length
30

L2 soil sample length - rest of core (in centimeters - cm) *
44

Total sample depth extracted (in centimetres - cm) *
74

Has a valid sample been extracted? *
A valid sample must be at least 30cm
Yes - more than 60cm extracted
Possibly - more than 30cm but less than 60cm extracted
No - <30cm or other issue

14. Capture the location of the soil core, ensuring that the GNSS device accuracy is < 1m. If the location being captured is for a non-sampled location, capture the location that the observation was made from.

15. The form will then progress through a series of required photos to be taken for each core:

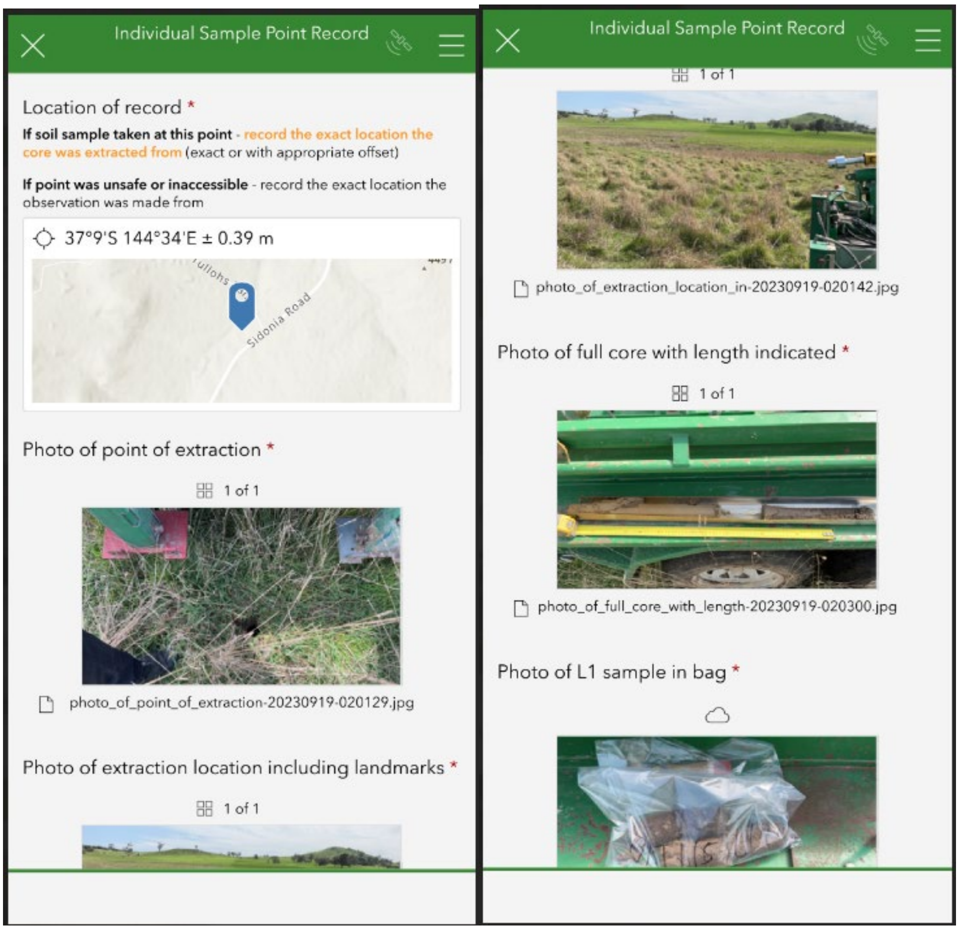
    a. *Photo of point of extraction*: photo showing the extraction point at ground surface.

b. *Photo of extraction location including landmarks*: photo showing the extraction point against a wider location shot, used by auditors to cross reference location during site visits.

c. *Photo of full core with length indicated*: photo showing the full core in the tube prior to bagging. This photo should clearly identify the top and bottom of the core and have a scale for reference.

d. *Photo of L1 sample in bag*: photo of the bagged L1 (0-30cm) sample clearly displaying the field code. **Note there is the option to scan in a barcode as the sample identifier at this step.**

e. *Photo of L2 sample in bag:* photo of the bagged L2 (30+ cm) sample clearly displaying the field code. **Note there is the option to scan in a barcode as the sample identifier at this step.**



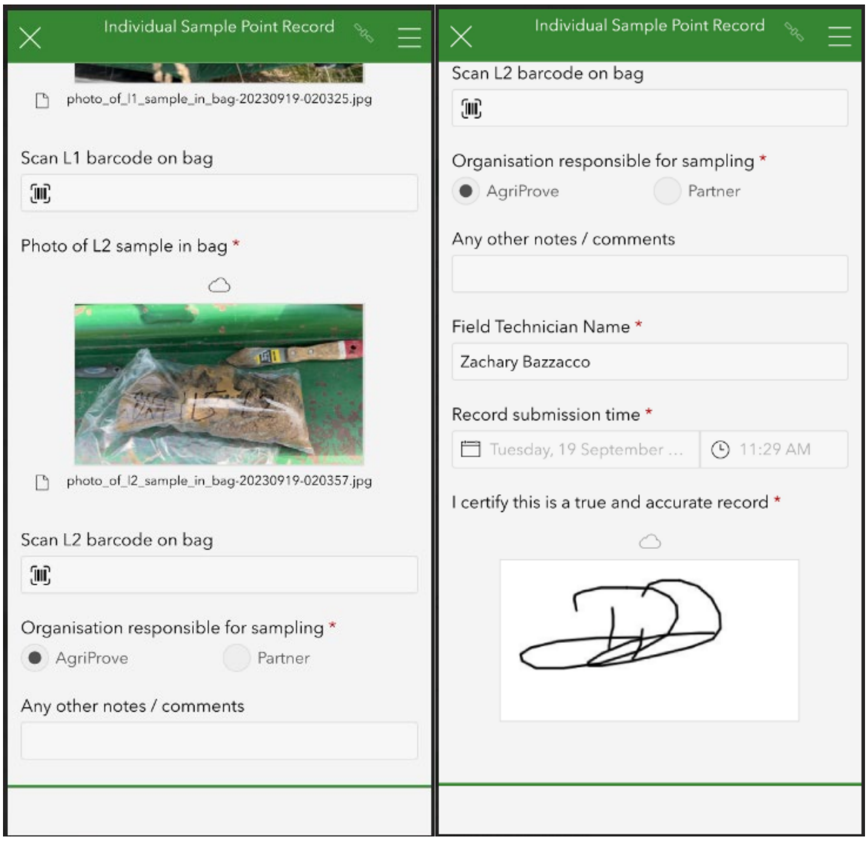
16. Under *Organisation responsible for sampling* select the most appropriate option (for example, AgriProve employees select *AgriProve,* third party contractors select *Partner.*

17. Add any notes or comments as required.

18. Add the name of the field technician responsible to *Field Technician Name.*

19. Record the submission time.

20. Add a signature confirming that the data collected for that core is a true and accurate record.



6.3 DIGITAL PROJECT AUDIT DATA REQUIREMENTS

AgriProve collects additional data while in field to facilitate fully digital project auditing. This data comprises an additional series of photographs and videos designed to satisfy audit testing of the accuracy of project mapping and to provide additional confirmation on the locations of soil core extractions. This data acquisition is to be completed using ArcGIS QuickCapture.

6.3.1 SETUP

1. Navigate to your app store and download ArcGIS QuickCapture.

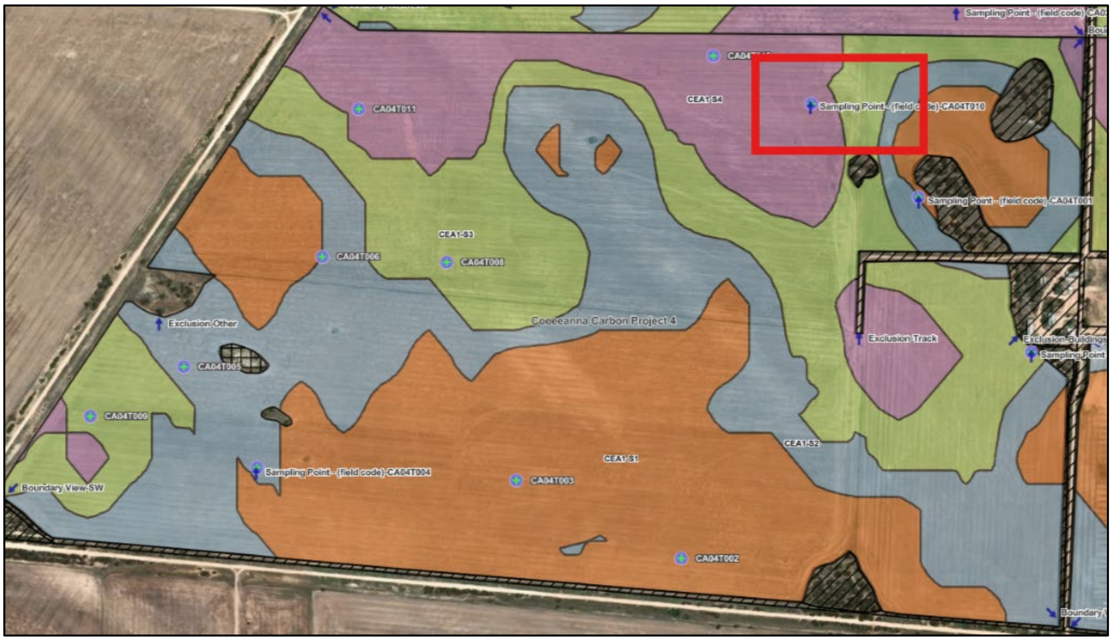
2. Login to QuickCapture using the same login credentials supplied by the GIS administrator above.

6.3.2 DATA COLLECTION

The locations of the required photos and videos are pre-determined by AgriProve. These cover three main categories:

1. Confirmation of project boundaries.
2. Confirmation of Exclusion Zones.
3. Video of the first core extracted from each project strata.
4. Photographic evidence of sample packing and boxing.

The locations of the required photos/videos for the exclusion zones, project boundaries, and soil sampling are pre-determined by AgriProve and will appear in Field Maps as arrows pointing in the direction of the required field of view of the photograph/video as shown below.

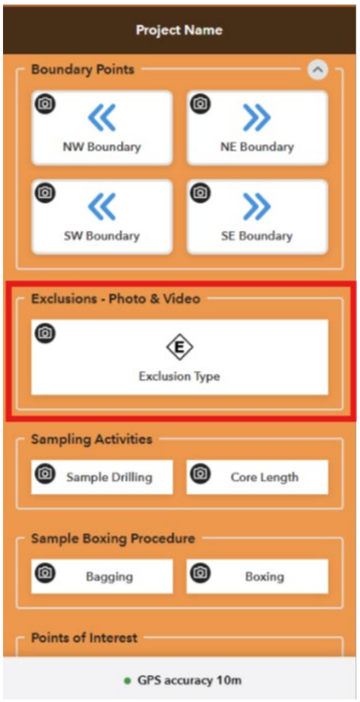


The process to acquire data for exclusion zones and project boundaries is as follows:

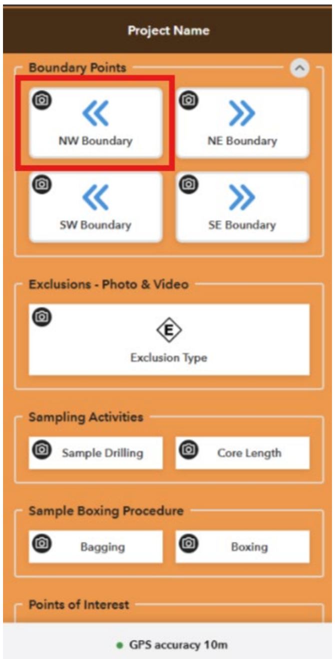
1. Navigate to the pre-determined location in Field Maps.

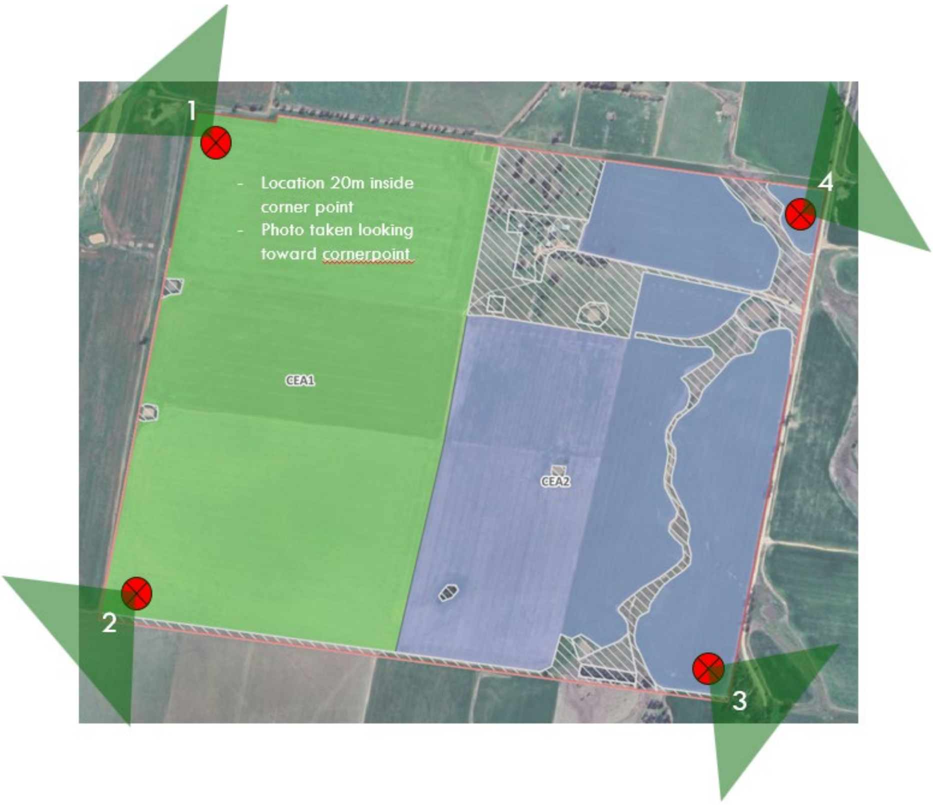
2. Open the QuickCapture app and enter the project name at the top of the screen.

3. For exclusion zones, click on the *Exclusion Type* button and select the corresponding Exclusion Type as referenced in the label in Field Maps, then take the photo of the Exclusion Zone. This photo should be taken approximately 10m away from the boundary of the relevant Exclusion Zone, with the view direction looking toward the excluded area. Use the widest possible camera angle to capture as much of the excluded area in shot as possible.



4. For project boundaries, click on the relevant boundary button depending on which location you have navigated to, then take the photo of the relevant project boundary. This photo should be taken approximately 20m away from the relevant boundary, with the view direction looking toward the boundary. Use the widest possible camera angle to capture as much of the project boundary in shot as possible.





5. For the first core extracted in each strata, collect a short video showing the location. QuickCapture only allows a 10 second video to be recorded. This video should be a 360 degree view of the location, taking in the location of the sampling rig. Click on the *Sample Drilling* button to acquire this video. **The Core Length button is not required**.

6. Finally use the *Bagging* button to take a photo of the storage of sample bags for the project, where possible with some field codes visible, and then use the *Boxing* button to take photos of the packed boxes prior to dispatch. Note that it is not necessary to be connected to GNSS GPS receivers for these photos, as they may be taken at a later time.

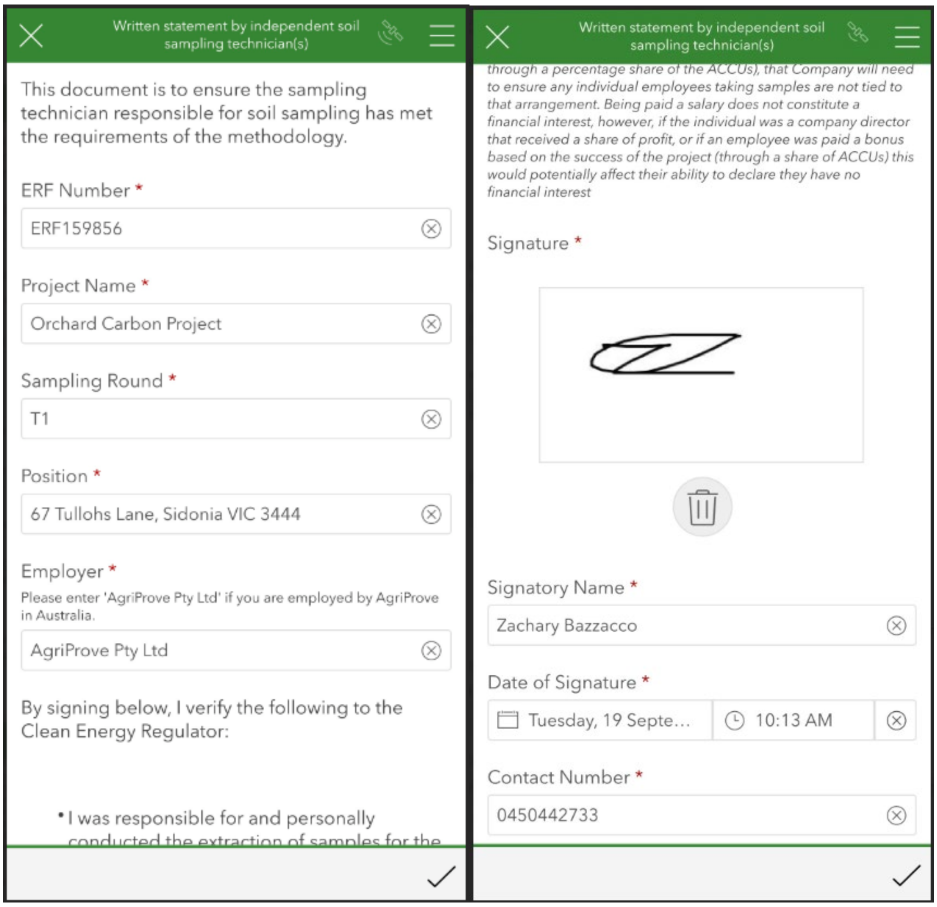
There is no requirement to sync data automatically as this will happen automatically once there is a data connection.

6.4 SAMPLING TECHNICIAN STATEMENT

Once all sampling for a given project area is complete, a written statement must be completed stating that sampling has been conducted in accordance with the method and supplement. To complete the field technician statement, within Field Maps, tap anywhere within the project strata on the screen.

A pop-up window will open with a link to the *Sampling Statement*. Tap on the link to open the sampling statement form. Enter all the required details into the form.

It is important that you read and understand the subsequent statement before signing at the bottom of the form. Once signed, tap the tick symbol to submit the form.



6.5 SYNC FIELD MAPS AND SURVEY 123

Once all sampling for a project has been completed, ensure that both Field Maps and Survey 123 have been synced using a stable data connection.

It is imperative that data is synced as soon as practical on completion of sampling (eg. as soon as a secure data connection is established) so that AgirProve's GIS Administrator can check that all data has been uploaded and is available prior to demobilisation so any data issues can be rectified at the time.

6.6 GENERAL REQUIREMENTS

While Field Maps and Survey 123 are designed to maintain compliance and data quality, there are a number of other general requirements that must be adhered to during sampling:

1. **A minimum of three samples per strata must be collected. This should be cross-checked prior to demobilisation.**

2. **Where used, reserve sample points must be taken in the same strata as the unused primary sample point.**

3. **If any obstruction (such as a tree or boulder) prevents sampling at the target location, the offsetting procedure can be employed by:**

    - **Moving north until the obstacle is cleared to a maximum of 4 metres.**
    - **If this does not clear the obstacle continue to change the direction of movement by 15 degree increments until the obstacle is cleared.**
    - **Ensure that in offsetting you do not cross strata boundaries.**
    - **Refer to *Section 1.2.b* of the Supplement.**

4. **At the completion of each core, it is imperative that the hole is filled in and closed prior to moving on to the next core. This can be done by collapsing or filling the top of the hole with a shovel.**

5. **All sample bags must be clearly labelled with the field code at a minimum. It is preferable that a barcoding system is used and that the barcode is firmly attached to each sample bag and protected such that it does not become obscured during transit. For clarity the field code comprises:**

    - **Project ID - Sample Round - Sequence (eg. OCT001)**

6. **All soil core measurements should be completed as accurately as possible, and it is imperative that all recovered soil for each layer is captured in the corresponding sample bag. Small errors in sample lengths and weights can cause large errors with calculations, particularly for bulk density.**

6.7 SAMPLE PACKING AND HANDLING

While completing sampling, where possible samples should be kept out of direct sunlight and stored in a secure manner to prevent damage to the bags. Wherever possible samples should also be kept cool.

All samples should be packed safely and securely in boxes for transport to the specified laboratory. Care should be taken to ensure samples will not move during transport and risk damaging sample bags and integrity.

Sample boxes should be no more than 20 kilograms in weight.

6.8 CHAIN OF CUSTODY

Chain of Custody (CoC) refers to the ability to show where a sample was located at every stage from extraction through to receipt at the relevant laboratory. There are two components to CoC comprising sample freight, and laboratory submittal documentation.

6.8.1 SAMPLE FREIGHT

Contractors are responsible for arranging freight of samples to the AgriProve specified laboratory. This should be complete using a reputable freight or courier service and that the Chain of Custody can be documented (for example through consignment notes or similar). It is imperative that freight of samples occurs as soon as possible after the completion of a sampling round (within 24 hours).

Contractors must provide AgriProve with freight details (consignment notes etc) once dispatched.

6.8.2 LABORATORY SUBMITTALS – MANUAL PROCESS

AgriProve will determine which laboratory the samples will be submitted to. Four labs are currently used for analysis:

- EAL
- HRL
- Nutrient Advantage
- CSBP

AgriProve will provide laboratory specific instructions to the contractor on engagement.

7 DAMAGE TO THIRD PARTY PROPERTY OR EQUIPMENT

If during the course of sampling operations any damage is incurred to landholder or other third-party equipment or property, the Head of Operations should be notified immediately of the type and extent of damage, and an incident report completed.

Where damage is incurred and AgriProve or third-party contractors are at-fault, AgriProve will compensate the affected party for repairs as required. This may result in an insurance claim being made dependent on the quantum and extent of costs associated with the damage.